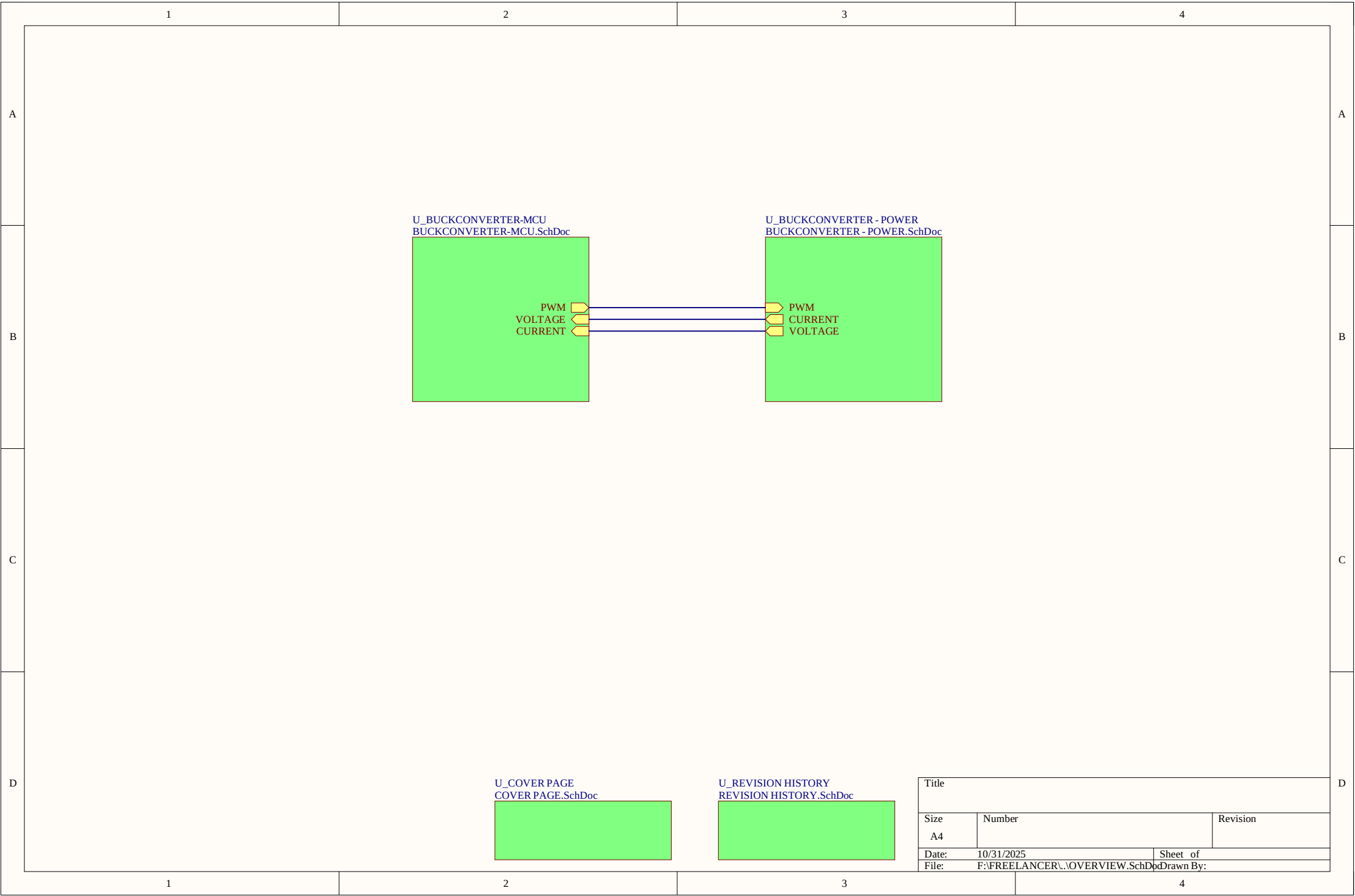




BUCK CONVERTER

31/10/2025

Variant: No variant

DRAFT 20-09-2025

Page	Index	Page	Index	Page	Index	Page	Index
1	OVERVIEW	11		21		31	
2	COVER PAGE	12		22		32	
3	BLOCK DIAGRAM	13		23		33	
4	BUCKCONVERTER - MCU	14		24		34	
5	BUCKCONVERTER - POWER	15		25		35	
6	HISTORY	16		26		36	
7		17		27		37	
8		18		28		38	
9		19		29		39	
10		20		30		40	

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational
design notes.

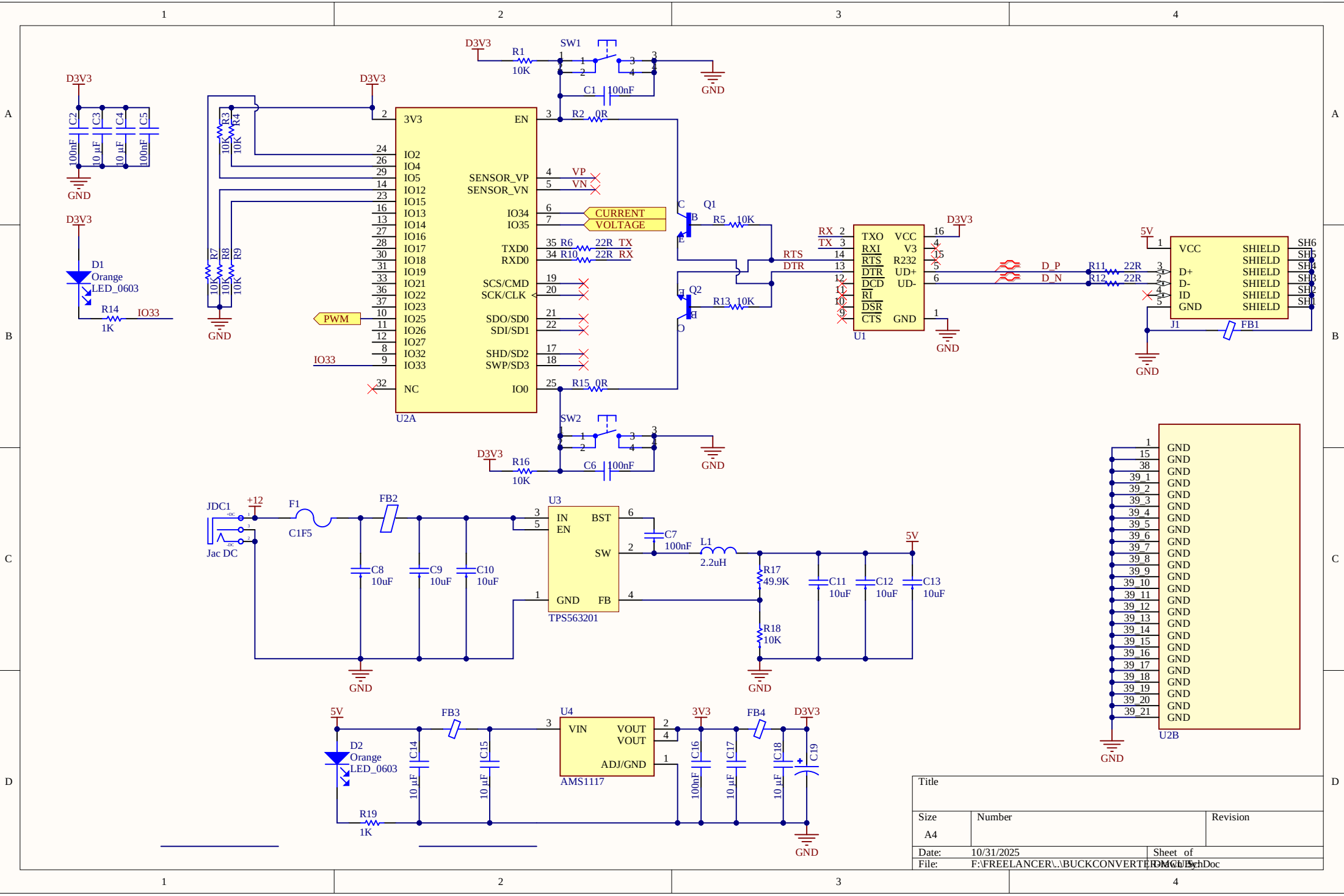
DESIGN NOTE:
Example text for cautionary
design notes.

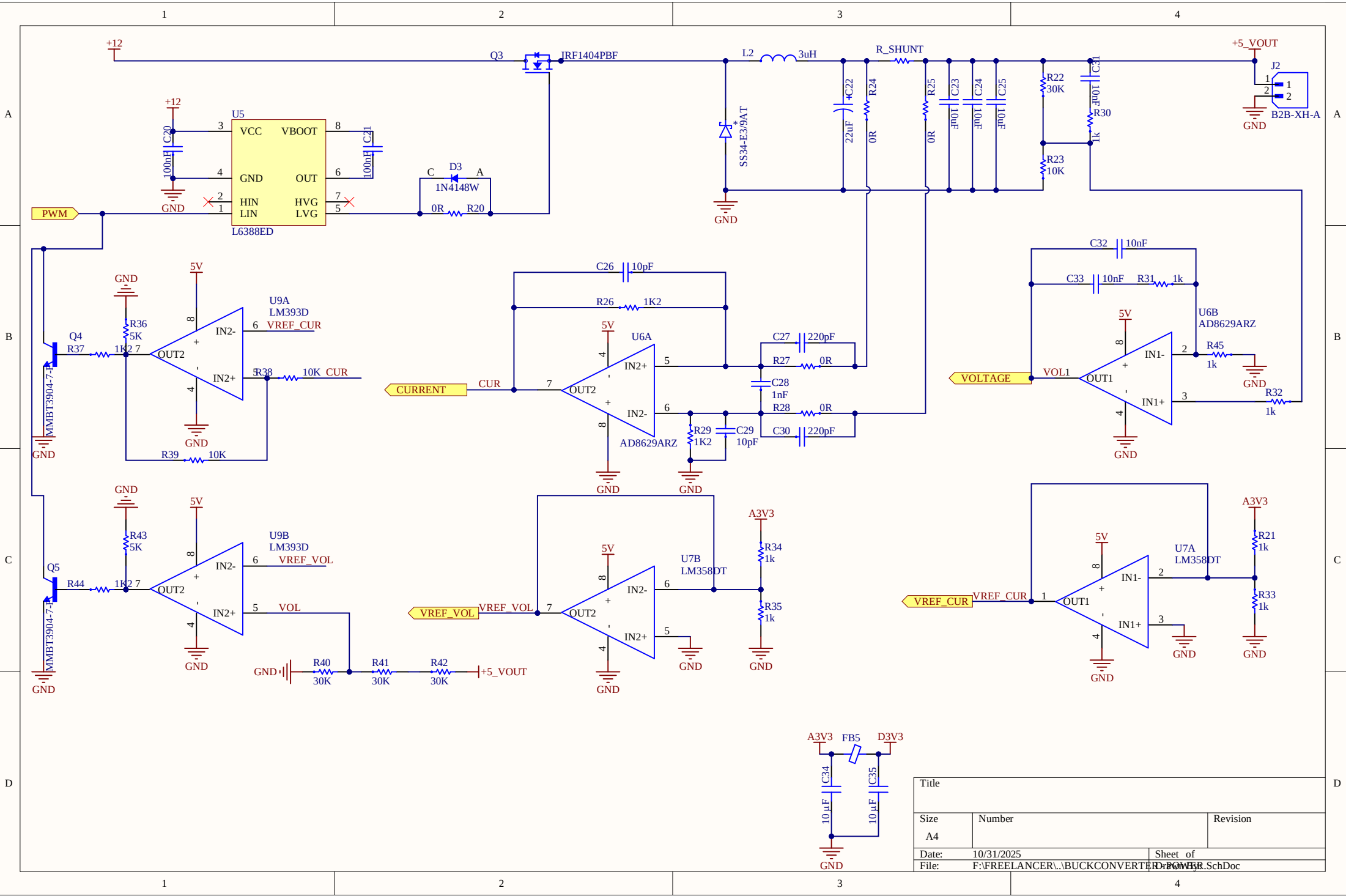
DESIGN NOTE:
Example text for debug notes.

DESIGN NOTE:
Example text for critical
design notes.

LAYOUT NOTE:
Example text for critical
layout guidelines.

Title		
Size A3	Number	Revision
Date: 10/31/2025	Sheet of	
File: F:\FREELANCER\COVER PAGE.Sch	Drawn By:	





Title		
Size	Number	Revision
A4		
Date:	10/31/2025	Sheet of
File:	F:\FREELANCER\BUCKCONVERTER.DrPower.SchDoc	10/31/2025

1	2	3	4
A			A
B			B
C			C
D			D
1	2	3	4

Title

Size

A4

Number

Revision

Date:

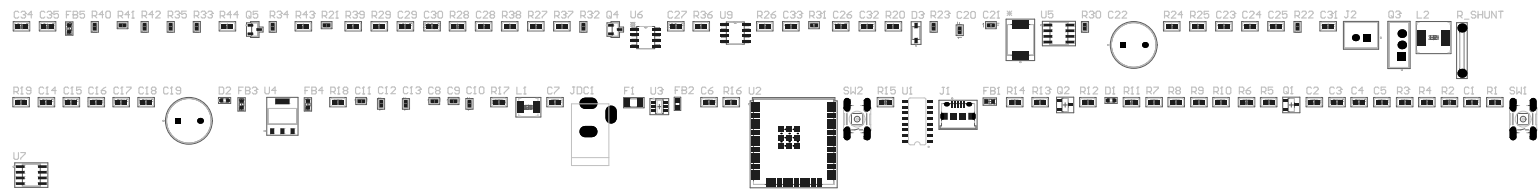
10/31/2025

Sheet of

File:

F:\FREELANCER\...\REVISION HISTORY\...

Drawn by:



Comment	Recognition	Designator	Footprint	Libref	Quantity
IS34-ES19AT	Rectifier Diode, Schottky, 1 Phase, 1 Element, 3A, 40V VFRM, Silicon, DO-144B		IS34	IS34-ES19AT	1
100pf	CAP0603 1TR 100P 10% 10V	C1, C2, C3, C6, C18	CAP 0603	04C107R100V100T	5
10pf	CAP0603 1TR 100P 10% 10V	C3, C4, C14, C15, C17, C18, C19, C25	CAP 0603	04C107R100V100T	6
100pf	CAP CER 1000P 10% 50V X7R 0603 SMD	C7	CAP-CER0603-0.0H	CAP-CER0603 1000P 10% 50V X7R 0603 SMD	1
10uF	CAP CER 100P 20% 6.3V X5R 0402 SMD	C8, C9, C10, C11, C12, C13	CAP-CER0402-20mH	CAP-CER0402 100P 20% 6.3V X5R 0402 SMD	6
ECADACH3331	Cap Aluminum 330uF 16V 20% (8 X 11.5mm) Radial Aluminum Cylindrical Can 3.5mm 360mA 1000V 1.2% Bulk	C19	330UF-5mm	ECADACH3331	1
1000pf	CAP CER 100P 20% 6.3V X5R 0402 SMD	C20, C21	CAP-CER0402-20mH	CAP-CER0402 100P 20% 6.3V X5R 0402 SMD	2
10uF		C22	100UF-5mm	CAP 45uF 63V 1	1
10uF	SMD Multilayer Ceramic Capacitor, 10 uF, 25 V, 0603 [1608 Metric], 1/4 20%, RFR	C23, C24, C25	Cap cer 0603	0RT188R16100M13 0	3
100f	CAP CER 2200P 10% 50V X7R 0603 SMD	C26, C29	CAP-CER0603-0.0H	CAP-CER0603 2200P 10% 50V X7R 0603 SMD	2
220pf	CAP CER 2200P 10% 50V X7R 0603 SMD	C27, C30	CAP-CER0603-0.0H	CAP-CER0603 2200P 10% 50V X7R 0603 SMD	2
1uF	CAP CER 2200P 10% 50V X7R 0603 SMD	C28	CAP-CER0603-0.0H	CAP-CER0603 2200P 10% 50V X7R 0603 SMD	1
10uF	SMD Multilayer Ceramic Capacitor, 10 uF, 25 V, 0603 [1608 Metric], 1/4 20%, RFR	C31, C32, C33	Cap cer 0603	0RT188R16100M13 0	3
Orange	LED SMD 0603 ORANGE 600NM 0603	C31, C32	LED	LED ORANGE 600NM 0603	2
1N4148W	Small Signal Diode, Single, 10V, 300 mA, 1.25 V, 4 Ns, 2 A Rats Component	C33	1N4148	1N4148W 7-P	1
C1P5	Fuse Chip Fast Acting 5A 125V SMD Solenoid Fuse 1206 1.1 x 1.5x1.1 0.03mm	F1	FLUSE C1P5	C1P5	1
330 Ohms @ 100 MHz	0805 1% RES 330 Ohm 0603 1LN	R61, R63, R64, R65	R6_0603	0LM188G33310H10	4
100MHz 2A 50mOhm	Permite Multi-Layer 1200mW 20% 1000Hz 2A 50mOhm 0603 0603 2-Pages 1.75	R62	R6 0603	0LM188PL1210H10	1
MICRO B		B1	MICRO USB B	MICRO USB B	1
020_0H-A		B2	020_0H-A	020_0H-A	1
04DC	dc DC Input	C01	C01	C01	1
1.2uH	IND MOI 100T 1.2uH 20% SA 0402 SMD	L1	IND 0402-2R2	INDMO0402S 100T 1.2uH 20% SA 0402 SMD	1
3uH	IND MOI 100T 1.5uH 20% SA 0402 SMD	L2	IND 06030-3uH	INDMO0603 100T 1.5uH 20% SA 0402 SMD	1
NMOS	MMBT3504 Series 40 V CE Breakdown 0.2 A NPN General Purpose Amplifier - SOT-23	Q1, Q2	3OT23-3	MMBT3504	2
0P140MRF	Single N-Channel 40 V 0.004 Ohm 136 nC 40VET* Power MOSFET - TO-252-3	Q3	TO220 AB	0P140MRF	1
MMBT3504-7-F	Transistor General Purpose BJT NPN 40 Volt 0.2A Automotive 3 Pin SOT-23	Q4, Q5	MMBT3504	MMBT3504-7-F	2
10K	Res Thick Film 0603 1.2K Ohm 5% 1/10W 0.0500pm/C Molded SMD SMD Paper 17R	R1, R3, R4, R5, R7, R8, R9, R13, R16	Res 0603	Resistor 2.2K	5
0R	Res Thick Film 0603 2.2K Ohm 5% 1/10W 0.0500pm/C Molded SMD SMD Paper 17R	R2, R15, R20	Res 0603	Resistor 2.2K, Resistor 0R	3
22R	Res Thick Film 0603 1.2K Ohm 5% 1/10W 0.0500pm/C Molded SMD SMD Paper 17R	R6, R10, R11, R12	Res 0603	Resistor 2.2K	4
1K	Res Thick Film 0603 2.2K Ohm 5% 1/10W 0.0500pm/C Molded SMD SMD Paper 17R	R14, R19	Res 0603	Resistor 2.2K	2
49.9K	RES CHIP 2K 1% 1/10W 0603 SMD	R17	RES-CHIP0603-0.49H-272	RES-CHIP0603 2K 1% 0603 SMD	1
10K	RES CHIP 2K 1% 1/10W 0603 SMD	R18, R38, R39	RES-CHIP0603-0.49H-272	RES-CHIP0603 2K 1% 0603 SMD	3
1u	RES CHIP 39K 1% 1/10W 0603 SMD	R21, R34, R31, R32, R38, R39, R40	RES-CHIP0402-0.39H	RES-CHIP0603 39K 1% 0603 SMD	8
10K	RES CHIP 39K 1% 1/10W 0402 SMD	R32, R40, R41, R42	RES-CHIP0402-0.39H	RES-CHIP0603 39K 1% 0402 SMD	4
10K	RES CHIP 39K 1% 1/10W 0402 SMD	R23	RES-CHIP0402-0.39H	RES-CHIP0603 39K 1% 0402 SMD	1
0R	RES CHIP 2K 1% 1/10W 0603 SMD	R24, R26, R27, R28	RES-CHIP0603-0.49H-272	RES-CHIP0603 2K 1% 0603 SMD	4
1K2	RES CHIP 2K 1% 1/10W 0603 SMD	R26, R28, R37, R41	RES-CHIP0603-0.49H-272	RES-CHIP0603 2K 1% 0603 SMD	4
5K	RES CHIP 2K 1% 1/10W 0603 SMD	R36, R43	RES-CHIP0603-0.49H-272	RES-CHIP0603 2K 1% 0603 SMD	2
0.01u	0.1uH/0.01	S_19A/01T	S_19A/01T	S_19A/01T	1
SW2	Switch Button 4pin	SW1, SW2	SWITCH	SW2	2
CS40C		U1		CS40C	1
ESP32-WROOM-32U6-N8	Multi-protocol Module ESP32-WROOM-32U6-ESP32-WROOM-V5-ESP32-01C0 V3	U2	ESP32S	ESP32-WROOM-32U6-N8	1
TP156AD001	RES DC/DC TP156AD001 3A SOT23-6 SMD	U3	SOT23-6	IC30C06001 600 TP156AD001 3A SOT23-6 SMD	1
AMS1117	3-A LOW DROPOUT VOLTAGE REGULATOR SOT-223	U4	AMS1117	AMS1117	1
LS88ED	Driver 2-CHT High and Low Side Half Bridge 6 Pin 10 N Tube I-C DRIVER H4-0 SIDEHV 8-SOIC	U5	SOIC 8	LS88ED	1
AD620ARZ	IC OPAMP ZERO-DRIFT CMOS BiCMOS Zero Drift Amplifier 2 Circuit Rail-to-Rail 0-SOIC	U6	AD620ARZ	AD620ARZ	1
LM595DT	Operational Amplifier, 2 Channels, 1.1 MHz, 14 V/Vs, 3V to 26V, 8 Pins, SOIC	U7	SOIC 8	LM595DT	1
LM595D	LM595 Series Dual 38 V 250 mA MT Low Power Voltage Comparator -SOIC-8	U9	LM593	LM595D	1